Environmental stochasticity drives mammals to increase their ranges

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- Average is not enough: environmental stochasticity drives home range expansion of mammals
- Moving beyond the average: environmental stochasticity as a driver of mammalian movement

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Abstract

The range size of an animal is often used of as a proxy for the animal's overall needs while also accounting for several latent variables, such as access to resources, vagility, competition, predation, and reproductive behaviour. Animals' average range sizes follow a strong power relationship with the animals' average mass and an exponent near 3/4, which suggests that average range sizes scale with animals' average metabolic needs. However, such a model only explains approximately 80% of the deviance in range size at the individual level. A great majority of the remaining 20% can be explained by accouning for inidividual-level variation in resource abundance and stochasticity (unpredictability). Using a large GPS dataset of 1599 terrestrial mammals, we present empirical evidence that the combination of mean and variance in NDVI explain causes them to expand their range, spend more time exploring, and move faster and more ballistically throughout the landscape. Our results suggest that growing environmental stochasticity will…

1 Introduction

like H = C / R paper[1], but: - R is now a random variable with mean and variation around the mean - Increased interest in environmental stochasticity has promoted thinking of env stoch (unpredictability) as a driver of animal movement (https://www.sciencedirect.com/science/article/pii/S0169534719302885) - variance in R is also important - relationship can now be nonlinear - because both factors are occurring at once and interacting, a multivariate approach is essential[2]

3

4

mass-only model explains ~80% of deviance

https://pubmed.ncbi.nlm.nih.gov/23020517/

- Migrating animals will have larger range sizes and smaller EDF values, and they will also be more susceptible to interannual variation. These two points work well together because they imply we should estimate changes in the mean as smooth changes that occur across years rather than among years, which means i should keep the k for years relatively low (e.g., (2025 - 1981) / 4 = 11 10)

wolverines have very low E(R), very large HR, and cache food

human activity is increasing stochasticity and conseuqnetly HR size, but it's also constraining movement capacities (Tucker paper) and attracting to urban traps

follow up after main figure: - effects of HFI - changes in variance over time: spatial needs will increase in the future, exacerbating issues - increases in variances => increases in movement rates => increases in energy expenditure, which increases HR size further? what if animals can't achieve an energetic balance? - increases in variances => increases in movement rates => increases in encounter rates => how will this affect trophic structures and predation rates? what if prey are forced to alter the predator/prey dynamics to try to

survive? - increases in env var can have strong cascading effects that then increase the var of other processes and lead to a series of strong cascades and maybe collapse

papers to use for structure: - jetz scaling of animal space use (https://pubmed.ncbi.nlm.nih.gov/15472074/) - tucker moving in the anthropocene (https://www.science.org/doi/10.1126/science.aam9712) - the search behavior of terrestrial mammals (https://www.biorxiv.org/content/10.1101/2022.12.31.521874v1)

artificial feeding can alter community structures in ways that are not reflected by NDVI: https://doi.org/10.1002/ecs2.70368

tapir results from first paper match the main figure! variance has u-shaped effect at low means: some variance at low means is beneficial because it allows to move across patches as they deplete, if animal can find good patches.

useful refs: - uses ndvi variance: https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecm.1348 -

2 Materials and Methods

for buffalo data, cite:

- W. M. Getz, S. Fortmann-Roe, P. C. Cross, A. J. Lyons, S. J. Ryan, C. C. Wilmers. LoCoH: Nonparameteric kernel methods for constructing home ranges and utilization distributions. PLoS ONE 2:2, e207 (2007).
- P. C. Cross, J. A. Bowers, C. T. Hay, J. Wolhuter, P. Buss, M. Hofmeyr, J. T. du Toit, W. M. Getz. Data from: Nonparameteric kernel methods for constructing home ranges and utilization distributions. Movebank Data Repository. DOI:10.5441/001/1.j900f88t (2016).

for coati data, cite:

- R. A. Powell, S. Ellwood, R. Kays. Stink or swim: techniques to meet the challenges for the study and conservation of small critters that hide, swim or climb and may otherwise

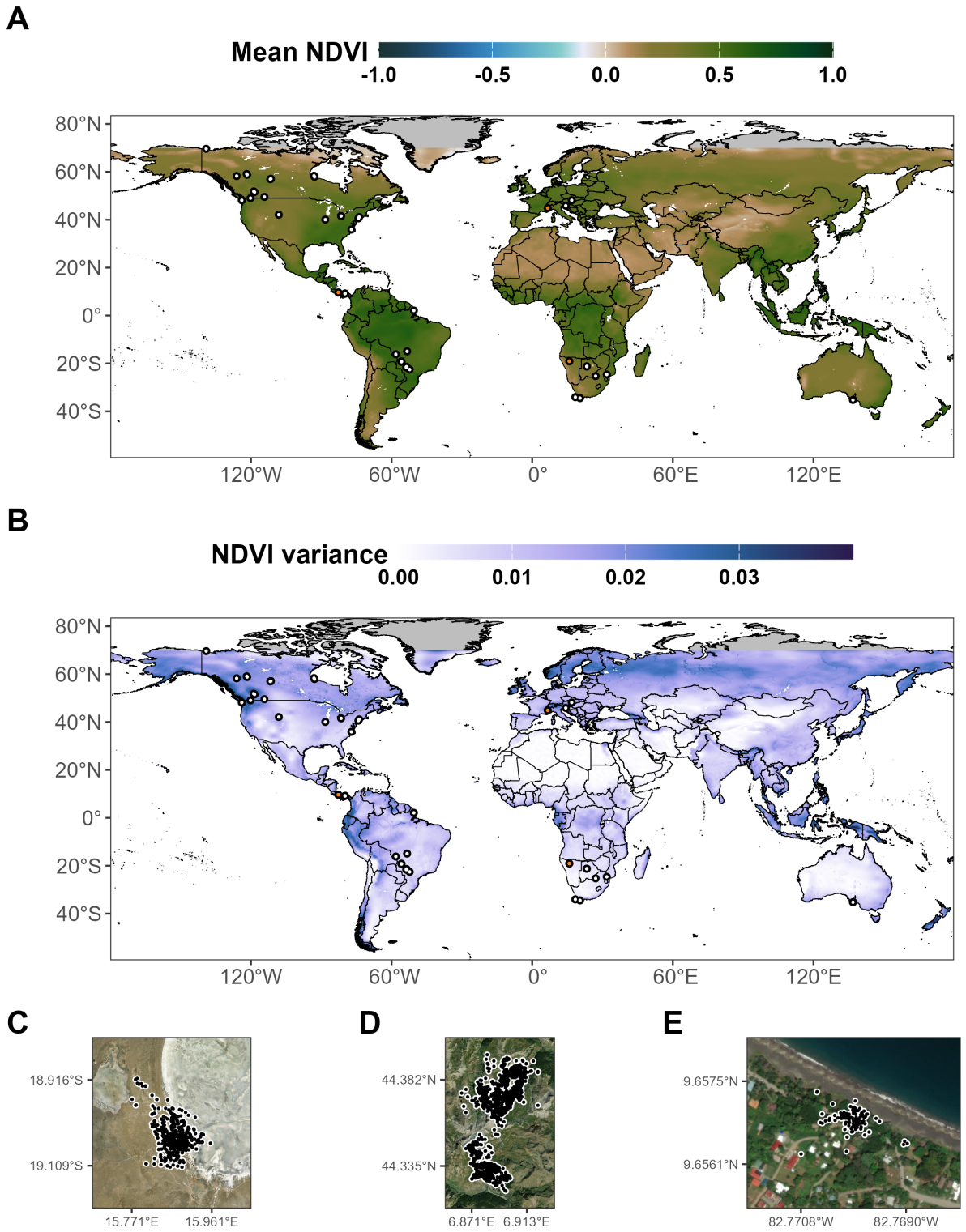


Figure 1: Maps of the rasters of mean NDVI (A) and variance in NDVI (B) used as proxies for mean and variance in resource abundance. The bottom three panels show examples of environments with: (C) low mean NDVI and variance (*Canis mesomelas* near the Etosha Pan in Namibia), (D) intermediate mean NDVI and variance (*Capra ibex* in the Alps), and (E) high mean NDVI and variance (*Choloepus hoffmanni* in Costa Rica).

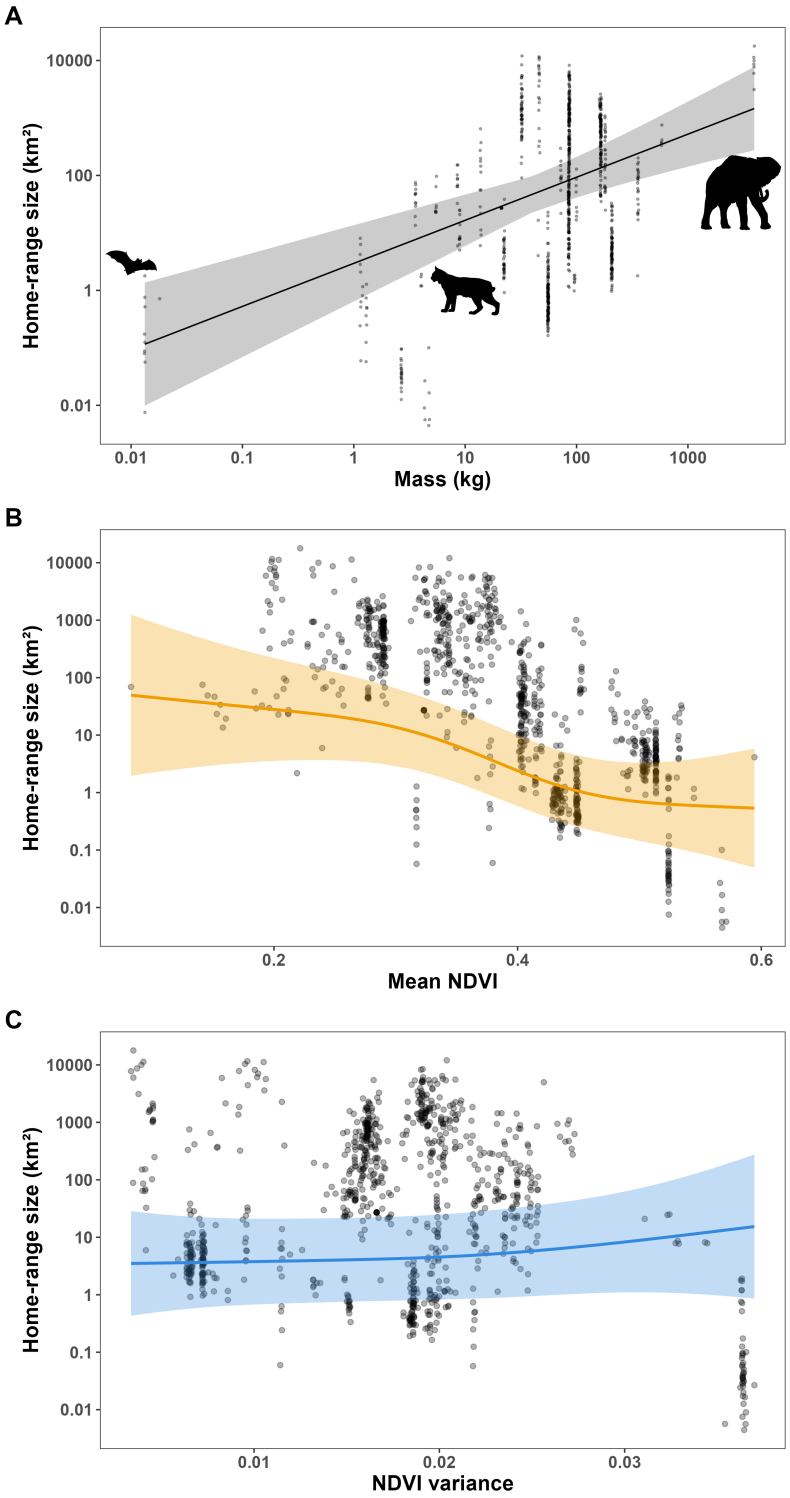


Figure 2: Partial effects of body size (A), mean NDVI (B), and NDVI variance (C) on home range size. Y axes are on the log scale. Masses in panel A are jittered horizontally to aid in visualization, but each species has a single value for mass. The animals in the bottom right corner of panel C have masses in the bottom 8th percentile of the dataset (species: *Bradypus variegatus*, *Carollia castanea*, *Carollia perspicillata*, *Choloepus hoffmanni*, *Dasyprocta punctata*, *Nasua narica*).

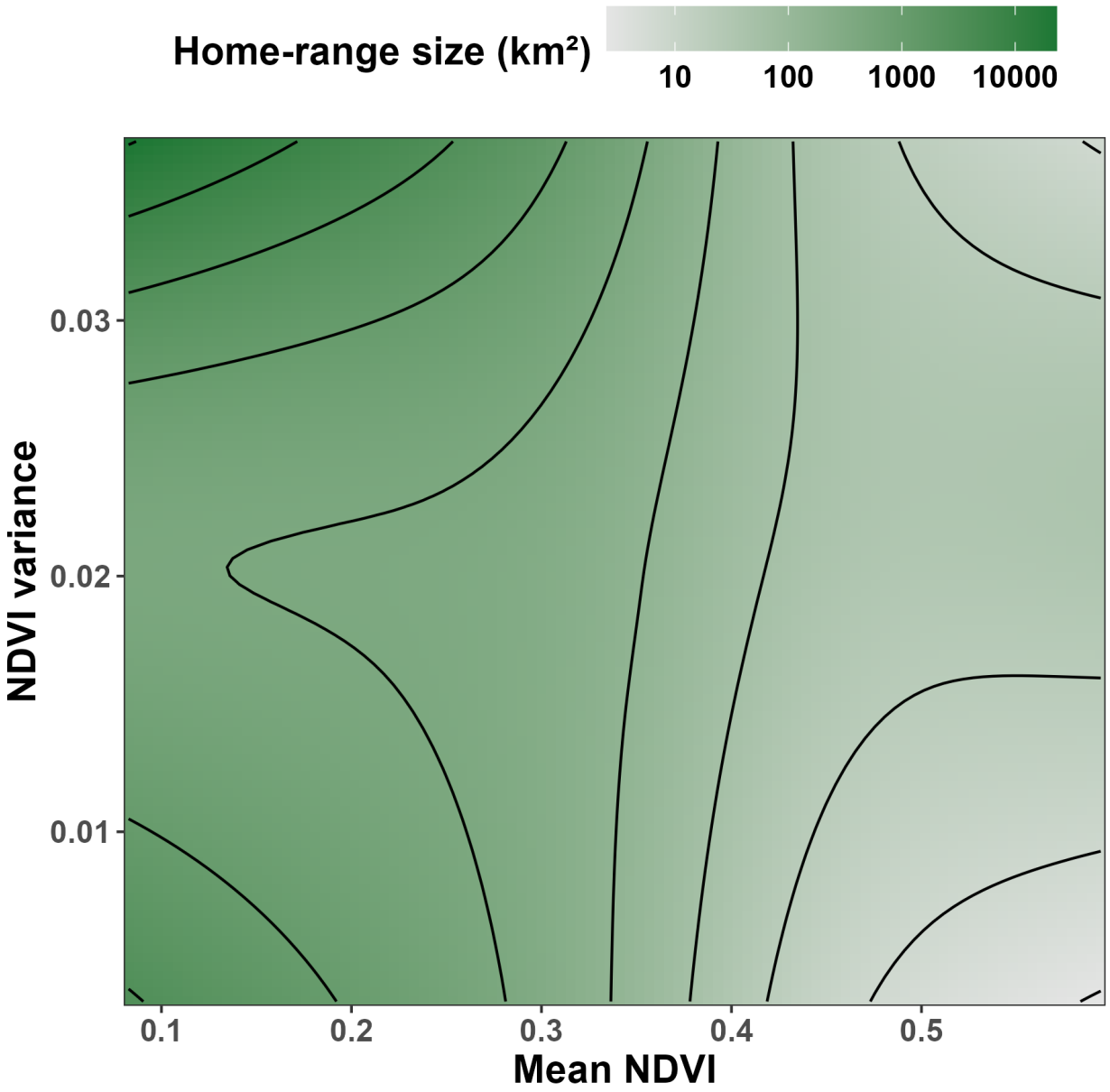


Figure 3: Interaction effects of mean NDVI and variance in NDVI on the home range size of a 85-kg mammal. Home range size is on the log scale. Contour lines indicate surface areas with identical home range size.

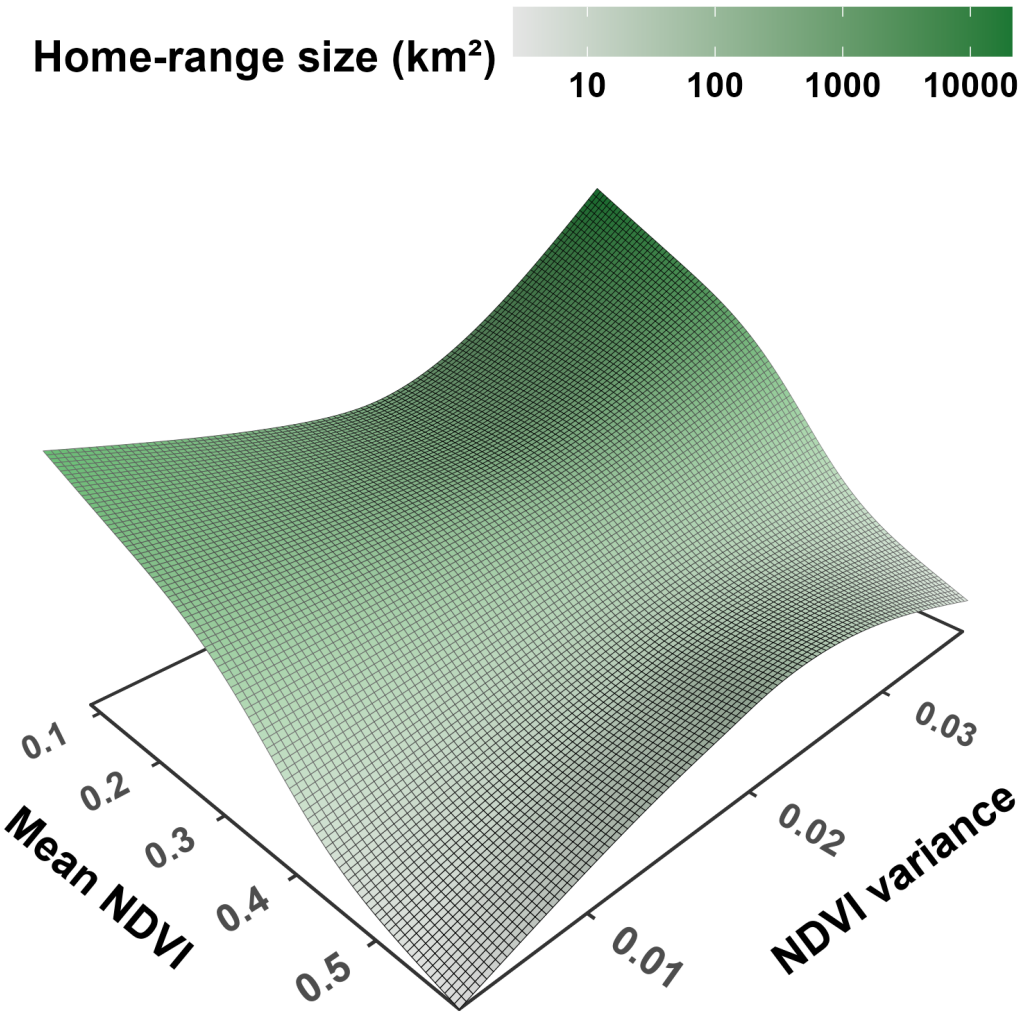


Figure 4: Interaction effects of mean NDVI and variance in NDVI on the home range size of a 85-kg mammal. Home range size is on the log scale in both the z axis and the fill scale.

make themselves unpleasant. In L. Harrington and D. W. Macdonald; Biology and Conservation of Mustelids and Procyonids (in preparation).

- R. Kays, B. T. Hirsch Data from: Stink or swim: techniques to meet the challenges for the study and conservation of small critters that hide, swim or climb and may otherwise make themselves unpleasant. Movebank Data Repository. DOI:10.5441/001/1.41076dq1 (2015).

for maned wolf, contact: Rogerio Cunha de Paula (rogercunha@gmail.com)

3 References and notes

1. Harestad, A. S. & Bunnel, F. L. Home Range and Body Weight–A Reevaluation. *Ecology* **60**, 389–402 (1979).

2. Polazzo, F. *et al.* Measuring the Response Diversity of Ecological Communities Experiencing Multifarious Environmental Change. *Global Change Biology* **30**, e17594 (2024).

3. Mezzini, S., Fleming, C. H., Medici, E. P. & Noonan, M. J. How resource abundance and resource stochasticity affect organisms' range sizes. *Movement Ecology* **13**, 20 (2025).

4. Brown, J. H., Gillooly, J. F., Allen, A. P., Savage, V. M. & West, G. B. TOWARD A METABOLIC THEORY OF ECOLOGY. *Ecology* **85**, 1771–1789 (2004).

4 Acknowledgments